

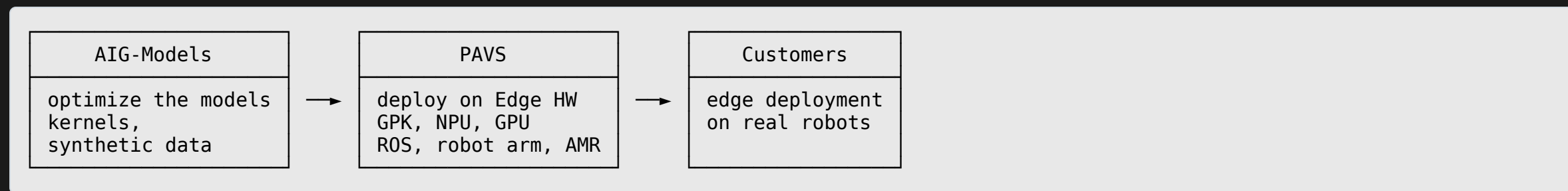
Meet & Greet with AIG-Models Team

Prateek Prabhanjan's team · **Hyperloom**

Their work: Hyperloom — an agentic system that optimizes kernels on AMD HW.

GEAK is the kernel-optimizing part of Hyperloom (other parts create traces, etc.). Today GEAK runs on MI300.

The two teams are two halves of one pipeline:



- **AIG-Models~(AIG-M)** optimize models — Kernel-Optimization, kernel generation, synthetic data Generation, Simulators.
- **PAVS** deployment of platforms & models — Edge HW (GPK, krk2e, PNR), PAI SDK, AMD-ROS, AMD-cUROBO, robot arm, AMR.

Purpose of this deck: the topics where the two teams overlap, and how we develop the collaboration.

Overlap Topics — and How We Develop Each

■ AIG-M (team) ■ People on the work ■ What AIG-M wants from us

Joint / technical work

1. Hyperloom → Edge HW (*Amol · Vaibhav · Karthik*)

- Extend GEAK (today on MI300) to GPK · krk2e · PNR
- Optimize PAI SDK models on GPU + NPU via GEAK
- An optimized NPU model → step toward an NPU-Hyperloom
- AIG-M to allocate an engineer to on-board us on GEAK

2. World Action Robotic Models (*Srini · Hari*)

- AIG-M developing & optimizing these now (trending soon)
- PAVS to understand, integrate & benchmark on Edge HW

3. Synthetic Data Generation (*Ravi · Kush · Hari*)

- AIG-M expertise in optimizing Genesis-style simulators
- PAVS robotics / sim-to-real can benefit

4. Kernel Generation

- AIG-M has proposed kernel-generation expertise
- We can explore an MLIR-style optimization library (as in MIGraphX) for NPU

What AIG-M wants from us

5. Assets Exchange — schedule a robotics session

- AIG-M wants to learn PAVS assets: AMD-ROS pipelines, AMD-cUROBO motion planner
- Setting up a walkthrough with the robotics team could help more collaboration and more AIG-Users for Robotics Team **PAVS-Robotics-DEV**
- They're interested in latest model development for edge deployment (**PAVS-AI-DEV**)

6. Customer Gateway — HW access

- AIG-M sees PAVS as their gateway to customers
- Share our Edge-HW to benchmark their models
- Run models on our robotic arm & AMR → Exposure feeds into AIG-M's R&D.